



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION SEMESTER II, SESSION 2020/2021

COURSE CODE	:	IFC 1014
COURSE NAME	:	CHEMISTRY
PROGRAMME	:	BRIDGING UTMSPACE
DURATION	:	3 HOURS
DATE	:	FEBRUARY 2022

INSTRUCTION TO CANDIDATES:

- 1.DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DOSO.
2. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

This examination question consists of (20) printed pages including this page.

$$q = ms\Delta T$$

$$= 225 \times 4.18 \times 14$$

$$= x$$

$$\Delta H = \frac{-x}{\frac{25}{40}} = -y \text{ J/mol}$$

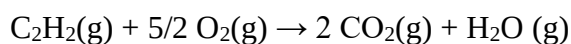
QUESTION 1 (10 MARKS)

1. 25 g of sodium (Na) is put into 200 mL of water. The reaction between the two produces sodium hydroxide (NaOH) and hydrogen gas, and increases the temperature by 14 °C.

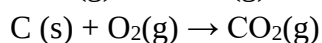
- Calculate the enthalpy change, ΔH , of the reaction.
($C_s = 4.18 \text{ J/}^\circ\text{C}\cdot\text{g}$; $d = 1 \text{ g/mL}$)
- Is this reaction endothermic or exothermic? *exo*
- Based on your calculation, determine what is considered as the surrounding in this experiment.
no heat loss to surrounding

(4 + 1 + 1 marks)

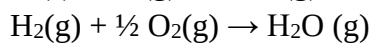
2. Given:



$$\Delta H = -1220.6 \text{ kJ/mol}$$

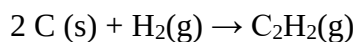


$$\Delta H = -503.5 \text{ kJ/mol}$$



$$\Delta H = -103.8 \text{ kJ/mol}$$

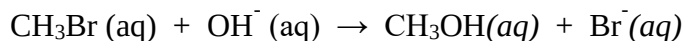
Calculate the enthalpy for the following reaction:



(4 marks)

QUESTION 2 (10 MARKS)

1. Consider the following reaction:



The rate law for this reaction is first order in CH_3Br and first order in OH^- . When $[\text{CH}_3\text{Br}]$ is $5.0 \times 10^{-3} \text{ M}$ and $[\text{OH}^-]$ is 0.120 M , the reaction rate at 303 K is 0.256 M/s .

- (a) Calculate the rate constant
(b) Determine the unit of the rate constant

(2 + 1 marks)

2. The gas phase decomposition of NO_2 , $2 \text{ NO}_2(\text{g}) \rightarrow 2 \text{ NO (g)} + \text{O}_2(\text{g})$, is studied at 383°C , giving the following data:

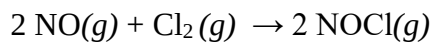
Time (s)	$[\text{NO}_2] \text{ (M)}$
0.0	0.100
5.0	0.017
10.0	0.0090
15.0	0.0062
20.0	0.0047

- (a) Determine the order of the reaction by using the mathematical method.
(b) Calculate the half life of the reaction.
(c) Do you expect the second half-life of the reaction to have the same value as the first one? Briefly explain your reasoning.

(4 + 1 + 2 marks)

QUESTION 3 (10 MARKS)

1. Nitric oxide (NO) reacts readily with chlorine gas as follows:



Given pressures of nitric oxide (NO) is 0.16 atm, chlorine (Cl₂) is 0.23 atm and NOCl is 0.56 atm.

- (a) Determine the equilibrium constant, K_p , of the reaction.
- (b) At 43 °C temperature, calculate the K_c of the reaction.
- (c) Determine the K_c for the reaction $5 \text{NOCl}(g) \rightarrow 5 \text{NO}(g) + 5/2 \text{Cl}_2(g)$

(2 + 2 + 2 marks)

2. Aurum (III) chloride, (AuCl₃), has a solubility of 54 g/100 mL of water. Determine the solubility-product constant, K_{sp} , of AuCl₃ in water.

(4 marks)

QUESTION 4 (15 MARKS)

1. When salt dissolves, it can form either acidic or basic salt solutions. Calcium chloride, CaCl_2 is a salt that is used to develop viscosity in polymers, food additives and preservatives, surfactants, and as a pH control in chemicals. On the other hand, sodium amide, NaNH_2 serves as a catalyst and a nucleophilic reagent in the industrial production of indigo dye.

- (a) The aqueous salt solution of CaCl_2 is neutral. Explain this statement using the salt dissociation equation in water.
- (b) The aqueous salt solution of NaNH_2 is basic. Explain this statement using the salt dissociation equation in water.

(3+3 marks)

2. During Christmas, Santa Claus found a bottle of used vinegar in his kitchen. Its chemical formula is CH_3COOH , also known as acetic acid. He wants to make a Caesar salad for his reindeer. In order to do that, you are requested to help Santa Claus to find the pH of the acetic acid in the 0.020 M solution. Given $K_a = 1.76 \times 10^{-5}$.

(5 marks)

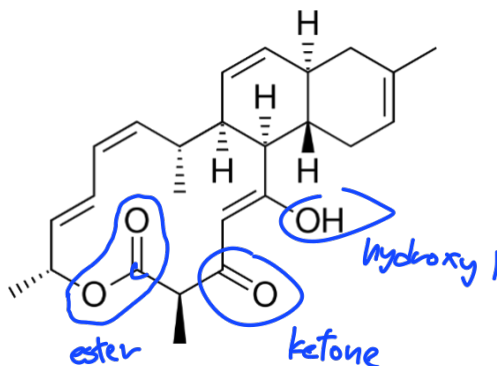
3. Buffer solutions are used by the human body to maintain a constant pH for biochemical processes such as enzyme activity. In addition, buffer is used in baby shampoo and soap to maintain the pH, thereby preventing rashes on the baby. Given an ethanoic acid, CH_3COOH with $K_a = 1.85 \times 10^{-5}$.

- (a) Calculate the pH of a buffer solution containing 0.150 mol/L ethanoic acid and 0.150 mol/L sodium ethanoate using the Henderson-Hasselbalch equation.
- (b) Give the proportion (ratio) for $[\text{CH}_3\text{COO}^-] : [\text{CH}_3\text{COOH}]$ with the same concentration that we should mix to produce a buffer solution of pH 5.00.

(1+3 marks)

QUESTION 5 (15 MARKS)

1. Anthracimycin is a new potential drug, derived from marine microorganism discovered in 2013. It is effective against anthrax and various other Gram-positive bacteria. Redraw the figure given below in your answer sheet and circle **FOUR** functional groups that you can find in the anthracimycin molecule.



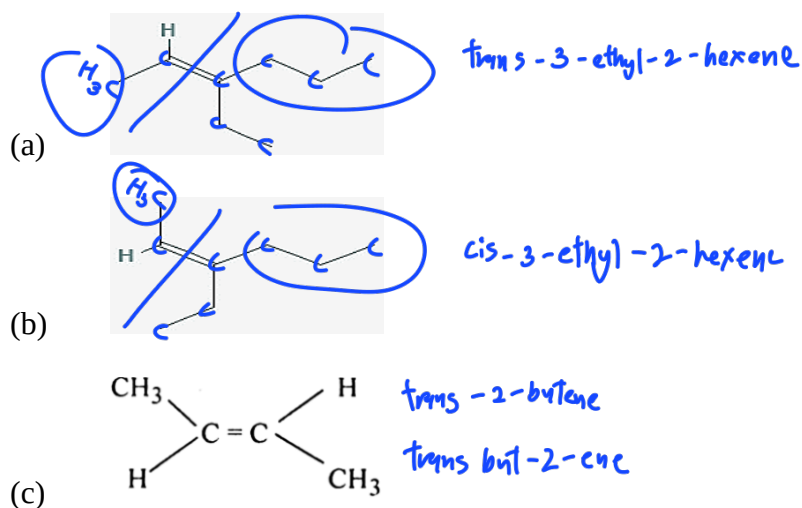
(4 marks)

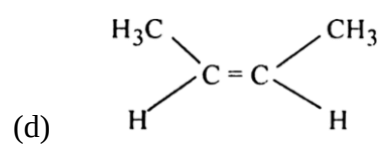
2. Draw the structural formula for the following molecules :

- (a) 1,2-dimethylcyclobutane
- (b) 2,2-dimethylpropane
- (c) 1,4-hexadiene
- (d) toluene
- (e) ortho-hydroxybenzoic acid
- (f) methylcyclopropane
- (g) propanal

(7 marks)

3. Write the IUPAC name for the given molecules below :





(4 marks)

1																	18
IA																	VIIIA
1	2											17	18				
H	He											F	Ne				
1.01	4.00											19.00	20.18				
3	4											9	10				
Li	Be											O	Ne				
6.94	9.01											16.00	20.18				
11	12											17	18				
Na	Mg											Cl	Ar				
22.99	24.31											35.45	39.95				
19	20											34	35				
K	Ca											Se	Kr				
39.1	40.08											78.96	83.80				
37	38											52	54				
Rb	Sr											Te	Xe				
85.47	87.62											127.6	131.29				
55	56											83	86				
Cs	Ba											Bi	Rn				
132.9	137.3											209	(222)				
87	88											(209)					
Fr	Ra																
(223)	(226)																

13	14	15	16	17	18
IIIA	IVA	VA	VIA	VIIA	
5	6	7	8	9	10
B	C	N	O	F	Ne
10.81	12.01	14.01	16.00	19.00	20.18
13	14	15	16	17	18
Al	Si	P	S	Cl	Ar
26.98	28.09	30.97	32.07	35.45	39.95
31	32	33	34	35	36
Ga	Ge	As	Se	Br	Kr
69.72	72.61	74.92	78.96	79.90	83.80
49	50	51	52	53	54
In	Sn	Sb	Te	I	Xe
114.82	118.71	121.76	127.6	126.9	131.29
81	82	83	84	85	86
Tl	Pb	Bi	Po	At	Rn
204.4	207.2	209	(209)	(210)	(222)

11	12	13	14	15	16	17	18
IB	IIIB	IIIA	IVB	VB	VIB	VIIA	
29	30	31	32	33	34	35	36
Cu	Zn	Ga	Ge	As	Se	Br	Kr
63.55	65.39	69.72	72.61	74.92	78.96	79.90	83.80
47	48	49	50	51	52	53	54
Ag	Cd	In	Sn	Sb	Te	I	Xe
107.87	112.41	114.82	118.71	121.76	127.6	126.9	131.29
79	80	81	82	83	84	85	86
Au	Hg	Tl	Pb	Bi	Po	At	Rn
197.0	200.6	204.4	207.2	209	(209)	(210)	(222)

65	66	67	68	69	70	71
Tb	Dy	Ho	Er	Tm	Yb	Lu
158.9	162.5	164.9	167.3	168.9	173.0	175.0
97	98	99	100	101	102	103
Bk	Cf	Es	Fm	Md	No	Lr
(247)	(251)	(252)	(257)	(258)	(259)	(260)

63	64	65	66	67	68	69	70	71
Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.0	175.0
95	96	97	98	99	100	101	102	103
Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
(243)	(247)	(247)	(251)	(252)	(257)	(258)	(259)	(260)

58	59	60	61	62	63	64	65	66	67	68	69	70	71
Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
140.1	140.9	144.2	(145)	150.4	152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.0	175.0
90	91	92	93	94	95	96	97	98	99	100	101	102	103
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
232.0	(231)	238.0	(237)	(244)	(243)	(247)	(247)	(251)	(252)	(257)	(258)	(259)	(260)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
IA	IIA	IIIA	IVB	VB	VIB	VIIA	VIIIA	VIIIA	VIIIA	IB	IIIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar
1.01	4.00	6.94	9.01	10.81	12.01	14.01	16.00	19.00	20.18	22.99	24.31	26.98	28.09	30.97	32.07	35.45	39.95
3	4	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
Li	Be	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca	Sc	Ti	V	Cr	Mn	Fe
6.94	9.01	22.99	24.31	26.98	28.09	30.97	32.07	35.45	39.95	39.1	40.08	44.96	47.88	50.94	52.00	54.94	55.85
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
85.47	87.62	88.91	91.22	92.91	95.94	(98)	101.07	102.91	106.42	107.87	112.41	114.82	118.71	121.76	127.6	126.9	131.29
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
Cs	Ba	La*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
132.9	137.3	138.9	178.5	180.9	183.9	186.2	190.2	192.2	195.1	197.0	200.6	204.4	207.2	209	(209)	(210)	(222)
87	88	89	104	105	106	107	108	109	110	111							
Fr	Ra	Ac^	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg							
(223)	(226)	(227)	(261)	(262)	(263)	(264)	(265)	(268)	(271)	(272)							

LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

FORMULAE

$$\frac{1}{\lambda_{\text{vac}}} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$4r = a\sqrt{2}$$

$$\text{rate} = \sqrt{\frac{1}{\text{density}}}$$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0592}{n} \log Q$$

STANDARD REDUCTION POTENTIAL ELECTROCHEMICAL SERIES

Half-Reaction	E (Volts)
$\text{K}^+ + \text{e}^- \rightleftharpoons \text{K}$	-2.924
$\text{Ba}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ba}$	-2.900
$\text{Ca}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ca}$	-2.760
$\text{Na}^+ + \text{e}^- \rightleftharpoons \text{Na}$	-2.712
$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$	-2.375
$\text{H}_2 + 2\text{e}^- \rightleftharpoons 2\text{H}^+$	-2.230
$\text{Al}^{3+} + 3\text{e}^- \rightleftharpoons \text{Al}$	-1.706
$\text{Mn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mn}$	-1.040
$\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$	-0.763
$\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$	-0.740
$\text{S} + 2\text{e}^- \rightleftharpoons \text{S}^{2-}$	-0.508
$2\text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{C}_2\text{O}_4$	-0.490
$\text{Cr}^{3+} + \text{e}^- \rightleftharpoons \text{Cr}^{2+}$	-0.410
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.409
$\text{Co}^{2+} + 2\text{e}^- \rightleftharpoons \text{Co}$	-0.280
$\text{Ni}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ni}$	-0.230
$\text{Sn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Sn}$	-0.136
$\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb}$	-0.126
$\text{Fe}^{3+} + 3\text{e}^- \rightleftharpoons \text{Fe}$	-0.036
$2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$	0.000
$\text{S}_4\text{O}_6^{2-} + 2\text{e}^- \rightleftharpoons 2\text{S}_2\text{O}_3^{2-}$	0.089
$\text{Sn}^{4+} + 2\text{e}^- \rightleftharpoons \text{Sn}^{2+}$	0.150
$\text{Cu}^{2+} + \text{e}^- \rightleftharpoons \text{Cu}^+$	0.158
$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	0.340
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightleftharpoons 4\text{OH}^-$	0.401
$\text{Cu}^+ + \text{e}^- \rightleftharpoons \text{Cu}$	0.522
$\text{I}_3^- + 2\text{e}^- \rightleftharpoons 3\text{I}^-$	0.534
$\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightleftharpoons \text{MnO}_2 + 4\text{OH}^-$	0.588
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{O}_2$	0.682
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	0.770
$\text{Hg}_2^{2+} + 2\text{e}^- \rightleftharpoons \text{Hg}$	0.796
$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	1.360